DO NOW — RECALL

# Linked lists with arrays and pointers

Name: _____ Class: _____ Date: _____

**1. What is the main difference between an array and a list?**

**2. In an array, what does an index identify?**

**3. In a linked list, what does null in a node's next pointer mean?**

**4. If the first item is at index 4 and its next pointer is 1, where is the second item?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

An array has a fixed size; a list can change size.

### 2. Model answer

The position used to access an array element.

### 3. Model answer

There is no next item.

### 4. Model answer

At index 1.

### By the end, you can…

• represent a linked list using parallel arrays and pointer indexes

• traverse a list by following pointers rather than adjacent indexes

• add and remove items by changing pointer values

### Vocabulary — know these terms

Recall the term — linked list

Recall the term — pointer

Recall the term — start pointer

Recall the term — next pointer

Recall the term — null

Recall the term — traverse

Recall the term — insert

Recall the term — remove

NEW LEARNING

# Read, annotate and explain

### The pointer chooses the next index

```
string[] data = {"Ben", "Daisy", "Alex", "Cara"};
int[] next = {3, -1, 0, 1};
int start = 2;
```

Run it: Alex → Ben → Cara → Daisy. The array positions stay fixed; the pointers create the logical order.

### Start pointer

The first logical item is stored at index 2, not necessarily index 0.

### Follow next pointers

Use -1 as null in code. Each pointer decides the next index.

### Insert at the front

The new item points to the old first item, then start changes.

### Remove after current

The pointer bypasses one node. The array data itself does not need to shift.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

## Traverse Alex → Ben → Cara → Daisy

## Use the start pointer

```
current = 2
data[2] = "Alex"
```

## Read next[2]

```
next[2] = 0
current = 0
data[0] = "Ben"
```

## Continue pointers

```
next[0] = 3
data[3] = "Cara"

next[3] = 1
data[1] = "Daisy"
```

## Reach null

```
next[1] = -1
stop
```

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY: Access the second linked-list item

OCR asks how the second item is reached using pointer values.

```
firstIndex = start
secondIndex = ?
secondData = data[?]
```

Method / working:

### Guided 2 — TRY: Traverse

Use the start pointer and follow next until -1.

```
int current = ?;
while (current != -1)
{
    Console.WriteLine(data[current]);
    current = ?;
}
```

Method / working:

### Guided 3 — TRY: Insert at front

newIndex is an unused array slot. Change the links, then traverse again.

```
next[newIndex] = ?;
start = ?;
```

Method / working:

### Guided 4 — TRY: Remove second item

Bypass the item immediately after start , then run the traversal to prove the change.

```
next[start] = ?;
```

Method / working:

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

### Question 1

Identify the first logical item
Run the starter. State the start index and first item. Explain why index 0 is not necessarily the head.
CODE:

```
using System;
public class Program
{
public static void Main()
{
string[] data = {"Ben", "Daisy", "Alex", "Cara", "", ""};
int[] next = {3, -1, 0, 1, -1, -1};
int start = 2; // slots 4 and 5 are unused
Console.WriteLine(data[start]);
}
}
```

### Question 2

Follow the next pointers
Use the unchanged starter. Write the sequence of indexes and names visited from start until -1.

### Question 3

Write the traversal loop
Replace the final print with a loop. Initialise current from start; output data[current]; follow next[current] until -1.
CODE:

```
current = start;
while (current != -1) { /* output, then follow link */ }
```

### Question 4

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1 · continued

Count reachable nodes
Add a counter to your traversal. Count reachable nodes, not data.Length, which includes unused capacity.
CODE:
int count = 0;
// increment once per reachable node

### Question 5

Insert after the head
Reset to the starter. Put Elliot in spare slot 4 and insert it immediately after Alex. Preserve the successor link before changing next[start].
CODE:
// First preserve the old successor in next[4].
// Then reconnect the predecessor.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

### Question 6

Insert a new first node
Reset to the starter. Store Zoe in unused slot 4. Link it to the old head, then update start. Traverse to verify.
CODE:
// New node -> old head; then move start.

### Question 7

Remove the second node
Reset to the starter. Bypass the second logical node by changing the head node's next pointer. Do not shift any data values.
CODE:
// Check the successor exists before bypassing it.

### Question 8

Remove the first node safely
Reset to the starter. If the list is nonempty, move start to next[start]. Explain what happens for a one-node list.
CODE:
if (start != -1) { /* update head */ }

### Question 9

Search by following links
Reset to the starter. Search for a name using current and next pointers. Report its array index or -1; test a present and absent name.
CODE:
// Follow links, not current++.

### Question 10

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2 · continued

Combine, test and explain
Combine traversal, insertion and deletion using the pointer arrays. Record expected and actual outputs for the tests below.
CODE:
Empty list: start=-1
One node: next[start]=-1
Insert into a spare slot
Delete head and middle nodes
Reject insertion if no slot is free
Use -1 only as the null marker. Reuse a removed slot only after marking it free in your chosen allocation scheme.

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**In an array-based linked list, the item at the next array index is not necessarily the next logical item.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

## GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

## Guided 1 — CHECK

Complete the same trace
CODE:
firstIndex = start
secondIndex = next[firstIndex]
secondData = data[secondIndex]

## Guided 2 — CHECK

Fill the same code
CODE:
int current = ?;
while (current != -1)
{
Console.WriteLine(data[current]);
current = ?;
}
CODE:
int current = start;
while (current != -1)
{
Console.WriteLine(data[current]);
current = next[current];
}

## Guided 3 — CHECK

Fill the same code
CODE:
next[newIndex] = ?;
start = ?;
CODE:
next[newIndex] = start;
start = newIndex;

## Guided 4 — CHECK

Fill the same code
CODE:
next[start] = ?;
CODE:
next[start] = next[next[start]];